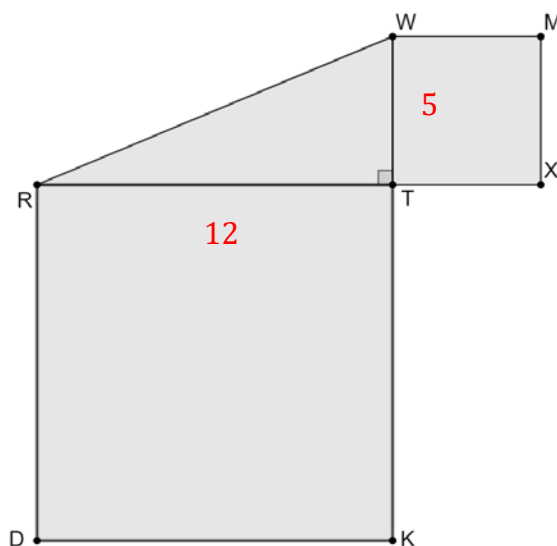


A diagram is shown where $RT = 12$ and $WT = 5$.



Complete the statements.

1. To show the Pythagorean theorem, a square that is drawn on the hypotenuse of $\triangle TWR$ will have an area equal to:

- ☐ $WT + RT$.
- ☐ $(WT + RT)^2$.
- ☒ the sum of the area of square $WMXT$ and the area of square $RTKD$.

2. The length of each side of the square that is drawn with side $\overline{RW}$ is **13**.

$$5^2 + 12^2 = c^2$$

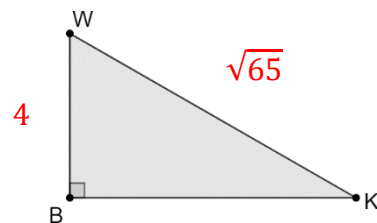
$$c = 13$$

3. $\triangle WKB$ is shown where $WK = \sqrt{65}$ and $WB = 4$. Find the exact length of $\overline{BK}$.

$$4^2 + b^2 = \sqrt{65}^2$$

$$b = 7$$

$BK = 7$

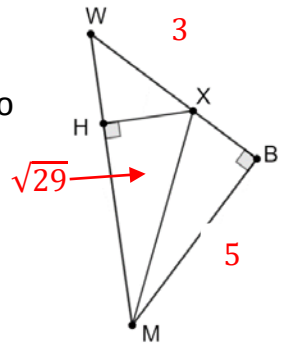


4. A right triangle has side lengths of $\sqrt{35}$ and $\sqrt{52}$. Find the exact length of the hypotenuse.

$$\sqrt{35}^2 + \sqrt{52}^2 = c^2$$

$$c = \sqrt{87}$$

5. Triangle WBM is shown where $m\angle WBM = 90^\circ$ and $m\angle XHM = 90^\circ$.
 $MX = \sqrt{29}$, $WX = 3$, and $BM = 5$.
 Determine the length of $\overline{WM}$. If necessary, round your answer to the nearest thousandth.



$$XB: a^2 + 5^2 = \sqrt{29}^2$$

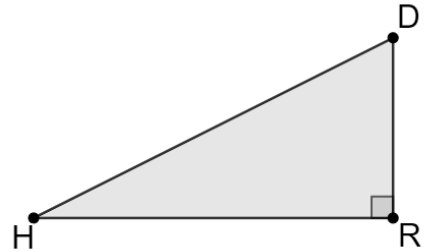
$$WM: 5^2 + 5^2 = c^2$$

$$a = 2$$

$$c = \boxed{WM = 7.071}$$

$$WB: 3 + 2 = 5$$

- $\triangle DRH$ is shown. Complete the table by finding the missing lengths. Round your answer to the nearest thousandth. Show your work below the table.



	Length of $\overline{DR}$	Length of $\overline{HR}$	Length of $\overline{DH}$
6.	28	15	31.765
7.	36	77	85
8.	17.493	$\sqrt{18}$	18
9.	33	56	65
10.	5	$\sqrt{7}$	$\sqrt{32}$

$$\begin{aligned} 6) 28^2 + 15^2 &= c^2 \\ c &= 31.765 \end{aligned}$$

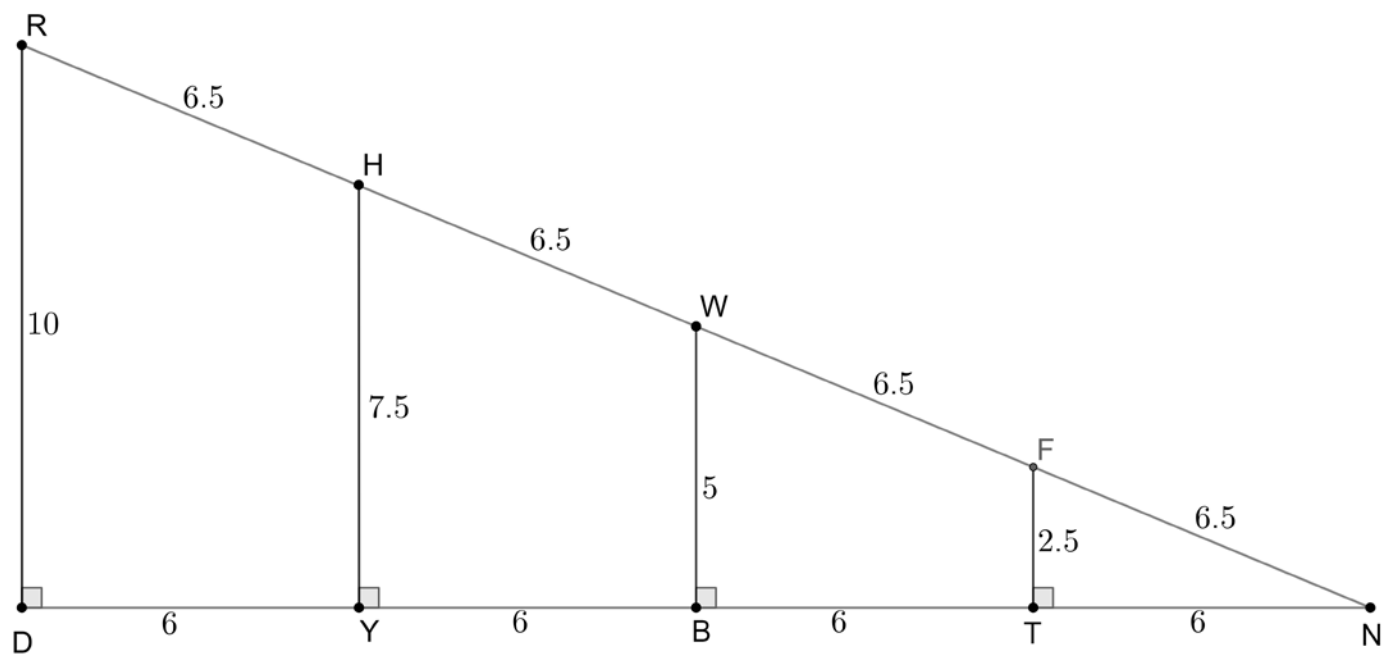
$$\begin{aligned} 9) 33^2 + b^2 &= 65^2 \\ b &= 56 \end{aligned}$$

$$\begin{aligned} 7) 36^2 + b^2 &= 85^2 \\ b &= 77 \end{aligned}$$

$$\begin{aligned} 10) a^2 + \sqrt{7}^2 &= \sqrt{32}^2 \\ a &= 5 \end{aligned}$$

$$\begin{aligned} 8) a^2 + \sqrt{18}^2 &= 18^2 \\ a &= 17.493 \end{aligned}$$

The diagram shows measurements for $\triangle RND$, $\triangle HNY$, $\triangle WNB$, and $\triangle FNT$.



Complete the table using $\angle N$ as the reference angle.

	Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
1.	$\triangle FNT$	$\frac{2.5}{6.5}$	$\frac{6}{6.5}$	$\frac{2.5}{6}$
2.	$\triangle WNB$	$\frac{5}{13}$	$\frac{12}{13}$	$\frac{5}{12}$
3.	$\triangle HNY$	$\frac{7.5}{19.5}$	$\frac{18}{19.5}$	$\frac{7.5}{18}$
4.	$\triangle RND$	$\frac{10}{26}$	$\frac{24}{26}$	$\frac{10}{24}$

For $\triangle FNT$, $\triangle WNB$, $\triangle HNY$, and $\triangle RND$ find the decimal equivalent for each ratio.

	Ratio	$\triangle FNT$	$\triangle WNB$	$\triangle HNY$	$\triangle RND$
5.	$\frac{\text{opposite}}{\text{hypotenuse}}$	0.385	0.385	0.385	0.385
6.	$\frac{\text{adjacent}}{\text{hypotenuse}}$	0.923	0.923	0.923	0.923
7.	$\frac{\text{opposite}}{\text{adjacent}}$	0.417	0.417	0.417	0.417

8. Complete the statements.

The decimal equivalents of the $\frac{\text{opposite}}{\text{hypotenuse}}$ ratios for triangles FNT , WNB , HNY , and

RND are ☐ equal.
☐ not equal.

The decimal equivalents of the $\frac{\text{adjacent}}{\text{hypotenuse}}$ ratios for

triangles FNT , WNB , HNY , and RND are ☐ equal.
☐ not equal.

The decimal equivalents

of the $\frac{\text{opposite}}{\text{adjacent}}$ ratios for triangles FNT , WNB , HNY , and RND are

☐ equal.
☐ not equal.

9. Complete the statement.

Since triangles FNT , WNB , HNY , and RND are

☐ acute
☐ obtuse
☐ right

triangles that share

☐ $\angle N$
☐ side $\overline{DN}$
☐ sides $\overline{DN}$ and $\overline{RN}$

the triangles are

☐ congruent.
☐ similar.

10. Complete the statement.

Because triangles FNT , WNB , HNY , and RND are

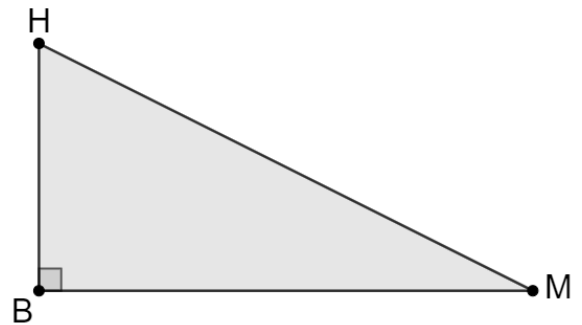
☐ congruent
☐ similar

their side lengths

are

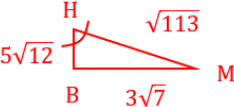
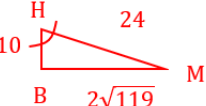
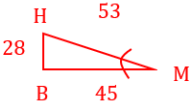
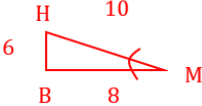
☐ congruent.
☐ proportional.

Triangle HBM is shown where $m\angle B = 90^\circ$.

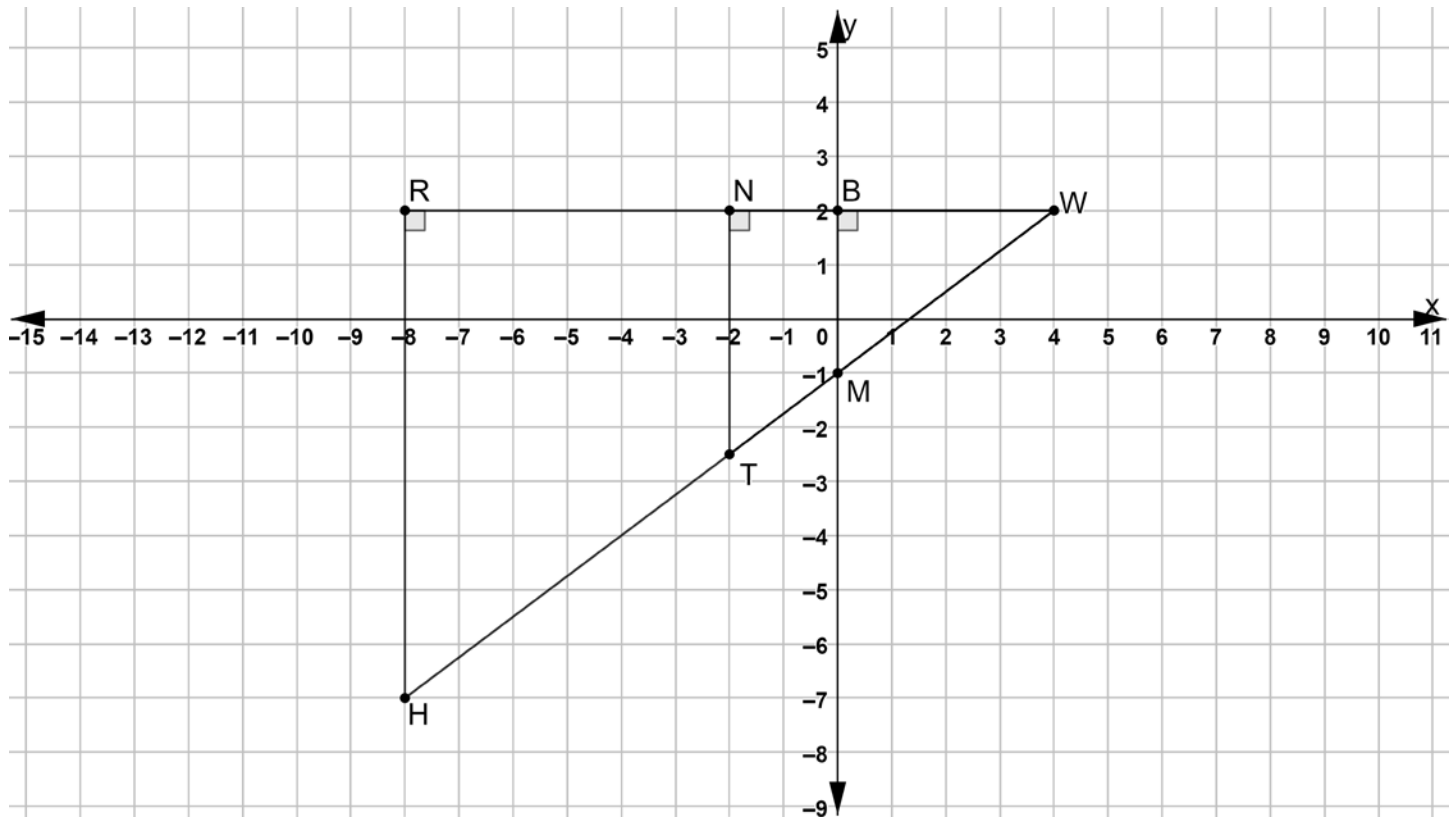


Complete the table using the given information.

	HB	BM	HM	Reference Angle	Cosine	Sine	Tangent
1.	8	15	17	$\angle M$ 	$\frac{A}{H} = \frac{15}{17}$	$\frac{O}{H} = \frac{8}{17}$	$\frac{O}{A} = \frac{8}{15}$
2.	8	15	17	$\angle H$ 	$\frac{A}{H} = \frac{8}{17}$	$\frac{O}{H} = \frac{15}{17}$	$\frac{O}{A} = \frac{15}{8}$
3.	33	56	65	$\angle M$ 	$\frac{A}{H} = \frac{56}{65}$	$\frac{O}{H} = \frac{33}{65}$	$\frac{O}{A} = \frac{33}{56}$
4.	$\sqrt{23}$	$\sqrt{17}$	$2\sqrt{10}$	$\angle M$ 	$\frac{A}{H} = \frac{\sqrt{17}}{2\sqrt{10}}$	$\frac{O}{H} = \frac{\sqrt{23}}{2\sqrt{10}}$	$\frac{O}{A} = \frac{\sqrt{23}}{\sqrt{17}}$
5.	$\sqrt{23}$	$\sqrt{17}$	$2\sqrt{10}$	$\angle H$ 	$\frac{A}{H} = \frac{\sqrt{23}}{2\sqrt{10}}$	$\frac{O}{H} = \frac{\sqrt{17}}{2\sqrt{10}}$	$\frac{O}{A} = \frac{\sqrt{17}}{\sqrt{23}}$

6.	$5\sqrt{2}$	$3\sqrt{7}$	$\sqrt{113}$	$\angle H$ 	$\frac{A}{H} = \frac{5\sqrt{2}}{\sqrt{113}}$	$\frac{O}{H} = \frac{3\sqrt{7}}{\sqrt{113}}$	$\frac{O}{A} = \frac{3\sqrt{7}}{5\sqrt{2}}$
7.	10	$2\sqrt{119}$	24	$\angle H$ 	$\frac{A}{H} = \frac{10}{24}$	$\frac{O}{H} = \frac{2\sqrt{119}}{24}$	$\frac{O}{A} = \frac{2\sqrt{119}}{10}$
8.	28	45	53	$\angle M$ 	$\frac{A}{H} = \frac{45}{53}$	$\frac{O}{H} = \frac{28}{53}$	$\frac{O}{A} = \frac{28}{45}$
9.	6	8	10	$\angle M$ 	$\frac{A}{H} = \frac{8}{10}$	$\frac{O}{H} = \frac{6}{10}$	$\frac{O}{A} = \frac{6}{8}$
10.	5	12	13	$\angle H$ 	$\frac{A}{H} = \frac{5}{13}$	$\frac{O}{H} = \frac{12}{13}$	$\frac{O}{A} = \frac{12}{5}$

Three right triangles are shown on the coordinate grid.



The coordinates of the vertices are shown.

$R(-8, 2)$

$N(-2, 2)$

$B(0, 2)$

$W(4, 2)$

$H(-8, -7)$

$T(-2, -2.5)$

$M(0, -1)$

1. Explain why these triangles are similar.

All three triangles share $\angle W$ and have a right angle

Additionally, all sides are proportional.

Complete the table.

	Triangle	Side	Length
2.	<i>BWM</i>	$\overline{BW}$	4
		$\overline{BM}$	3
		$\overline{MW}$	5 $4^2 + 3^2 = c^2$
3.	<i>NWT</i>	$\overline{NW}$	6
		$\overline{TN}$	4.5
		$\overline{TW}$	7.5 $6^2 + 4.5^2 = c^2$
4.	<i>RWH</i>	$\overline{RW}$	12
		$\overline{RH}$	9
		$\overline{HW}$	15 $12^2 + 9^2 = c^2$

Using $\angle W$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

	Triangle	Cosine	Sine	Tangent
5.	<i>BWM</i>	$\frac{A}{H} = \frac{4}{5} = 0.8$	$\frac{O}{H} = \frac{3}{5} = 0.6$	$\frac{O}{A} = \frac{3}{4} = 0.75$
6.	<i>NWT</i>	$\frac{A}{H} = \frac{6}{7.5} = 0.8$	$\frac{O}{H} = \frac{4.5}{7.5} = 0.6$	$\frac{O}{A} = \frac{4.5}{6} = 0.75$
7.	<i>RWH</i>	$\frac{A}{H} = \frac{12}{15} = 0.8$	$\frac{O}{H} = \frac{9}{15} = 0.6$	$\frac{O}{A} = \frac{9}{12} = 0.75$

Use your calculator to find each. Round to the nearest thousandth.

	Trigonometric Ratio	Value rounded to nearest thousandth
8.	$\cos(37^\circ)$	0.799
9.	$\sin(37^\circ)$	0.602
10.	$\tan(37^\circ)$	0.754

Geometry

Unit 7

Lesson 5 Homework

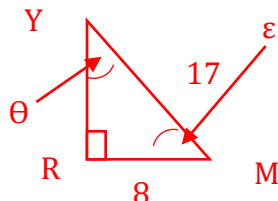
Name Answer Key

Date _____

Period _____

$\triangle YRM$ is a right triangle where $m\angle YRM = 90^\circ$, $RM = 8$, and $YM = 17$.

1. Determine $m\angle YMR$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.



$$\sin(\theta) = \frac{\text{opp.}}{\text{hyp.}} = \frac{8}{17}$$

$$\theta = \sin^{-1}\left(\frac{8}{17}\right) = 28.07^\circ$$

$$90^\circ + \theta + \varepsilon = 180^\circ$$

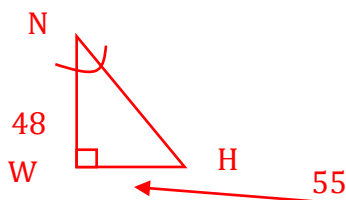
$$\therefore \varepsilon = 180^\circ - (90^\circ + 28.07^\circ)$$

$$= 61.93^\circ$$

$$m\angle YMR = 61.928^\circ$$

$\triangle WHN$ is a right triangle where $m\angle NWH = 90^\circ$, $NW = 48$, and $WH = 55$.

2. Determine $m\angle WNH$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

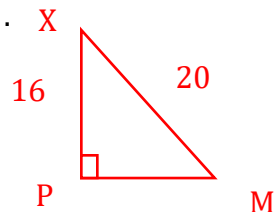


$$\tan \theta = \frac{55}{48}$$

$$\tan^{-1}\left(\frac{55}{48}\right) = 48.888^\circ$$

$\triangle XPM$ is a right triangle where $m\angle XPM = 90^\circ$, $XP = 16$, and $XM = 20$.

3. Determine the length of $\overline{MP}$. If necessary, round your answer to the nearest thousandth.



$$16^2 + b^2 = 20^2$$

$$256 + b^2 = 400$$

$$b^2 = 144$$

$$b = 12 \quad \overline{MP} = 12$$

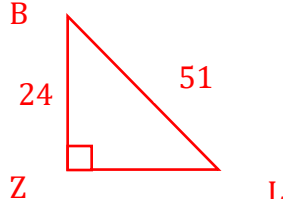
4. Determine $m\angle PMX$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{16}{12}$$

$$\tan^{-1}\left(\frac{16}{12}\right) = 53.130^\circ$$

$\triangle LBZ$ is a right triangle where $m\angle BZL = 90^\circ$, $BZ = 24$, and $LB = 51$.

5. Determine the length of $\overline{ZL}$. If necessary, round your answer to the nearest thousandth.



$$24^2 + b^2 = 51^2$$

$$576 + b^2 = 2601$$

$$b^2 = 2025$$

$$b = 45 \quad \overline{ZL} = 45$$

6. Determine $m\angle LBZ$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{45}{24}$$

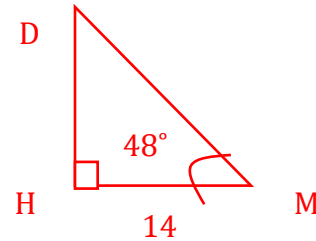
$$\tan^{-1}\left(\frac{45}{24}\right) = 61.928^\circ$$

$\triangle MDH$ is a right triangle where $m\angle DHM = 90^\circ$, $m\angle DMH = 48^\circ$, and $HM = 14$.

7. Determine the length of $\overline{DH}$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan(48^\circ) = \frac{\overline{DH}}{14}$$

$$\therefore \overline{DH} = 14 \cdot \tan(48^\circ) = \boxed{15.549}$$

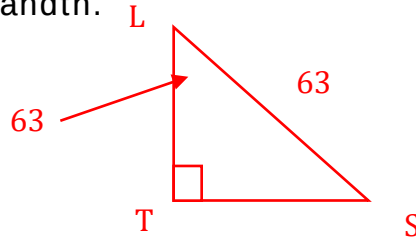


8. Explain why neither the cosine nor the sine ratio could be used to determine the length of $\overline{DH}$.

In order to use sine or cosine rate, the length of the hypotenuse of the right triangle must be given. Since the length of the hypotenuse is not provided, the only ratio that is possible to use based on the given information is tangent.

$\triangle TLS$ is a right triangle where $m\angle LTS = 90^\circ$, $m\angle TLS = 63^\circ$, and $LS = 63$.

9. Determine the length of $\overline{TL}$. Show your work. If necessary, round your answer to the nearest thousandth.



$$\cos(63) = \frac{x}{63}$$

$$28.601 = x$$

10. Determine the length of $\overline{TS}$. Show your work. If necessary, round your answer to the nearest thousandth.

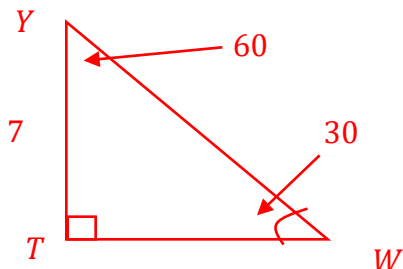
$$28.601^2 + x^2 = 63^2$$

$$x^2 = 3150.983$$

$$x = 56.134$$

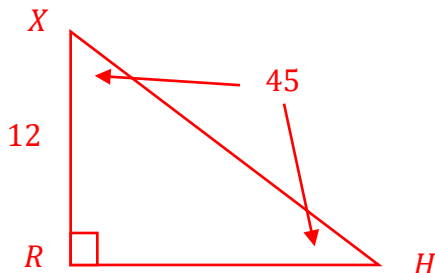
$\triangle YWT$ is a right triangle where $m\angle YTW = 90^\circ$, $m\angle YWT = 30^\circ$, and $YT = 7$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
1.	$\overline{WT}$	$7\sqrt{3}$
2.	$\overline{YW}$	14



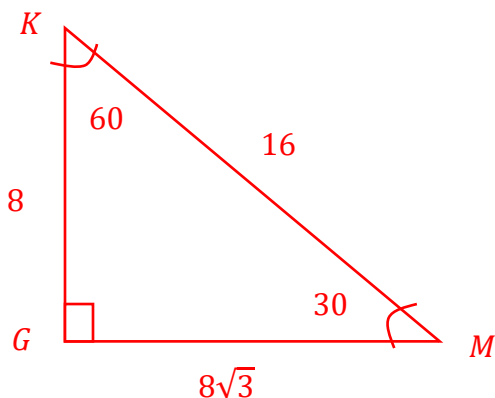
$\triangle RXH$ is a right triangle where $m\angle XRH = 90^\circ$, $m\angle RXH = 45^\circ$, and $XR = 12$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
3.	$\overline{RH}$	12
4.	$\overline{XH}$	$12\sqrt{2}$



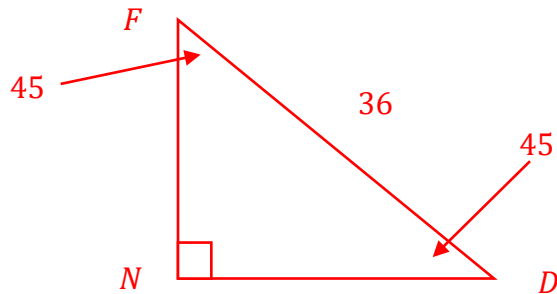
$\triangle KGM$ is a right triangle where $m\angle MGK = 90^\circ$, $m\angle GKM = 60^\circ$, and $KM = 16$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
5.	$\overline{KG}$	8
6.	$\overline{GM}$	$8\sqrt{3}$



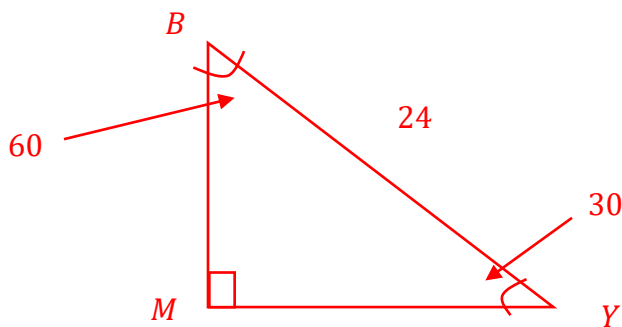
$\triangle FND$ is a right triangle where $m\angle FND = 90^\circ$, $m\angle NDF = 45^\circ$, and $DF = 36$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
7.	$\overline{ND}$	$18\sqrt{2}$
8.	$\overline{NF}$	$18\sqrt{2}$

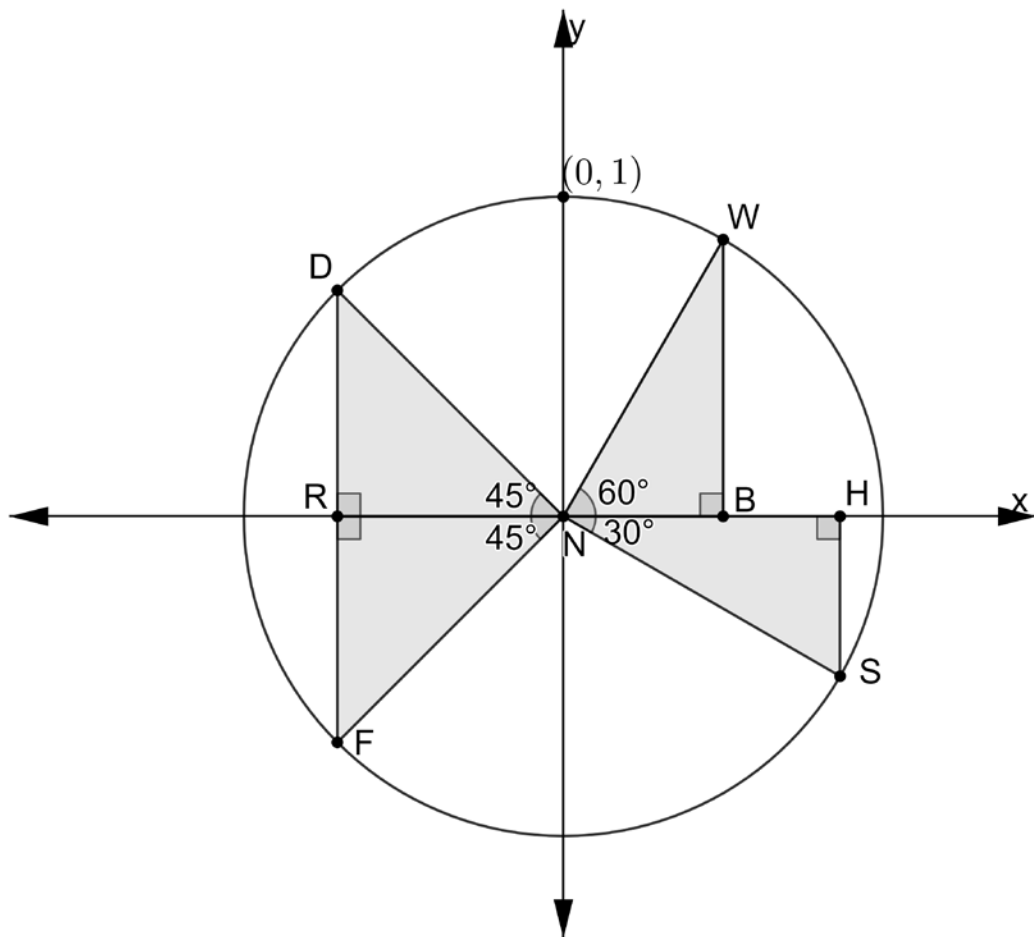


$\triangle YBM$ is a right triangle where $m\angle YMB = 90^\circ$, $m\angle MYB = 30^\circ$, and $YB = 24$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
9.	$\overline{MB}$	12
10.	$\overline{YM}$	$12\sqrt{3}$



Circle N and right triangles NWB , NHS , DNR , and RFN are shown on the coordinate plane.



1. What is the length of the radius of circle N ?

1

Complete the table.

	Segment	Length
2.	$\overline{NW}$	1
3.	$\overline{DN}$	1

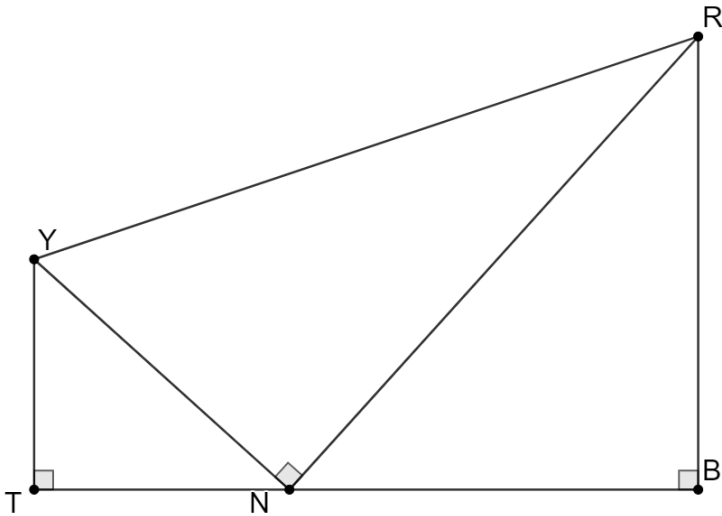
Complete the table.

	Point	Coordinates
4.	B	$\left(\frac{1}{2}, 0\right)$
5.	H	$\left(\frac{\sqrt{3}}{2}, 0\right)$
6.	R	$\left(\frac{-\sqrt{2}}{2}, 0\right)$
7.	W	$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
8.	D	$\left(\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
9.	F	$\left(\frac{-\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right)$
10.	S	$\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

Geometry
Unit 7
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

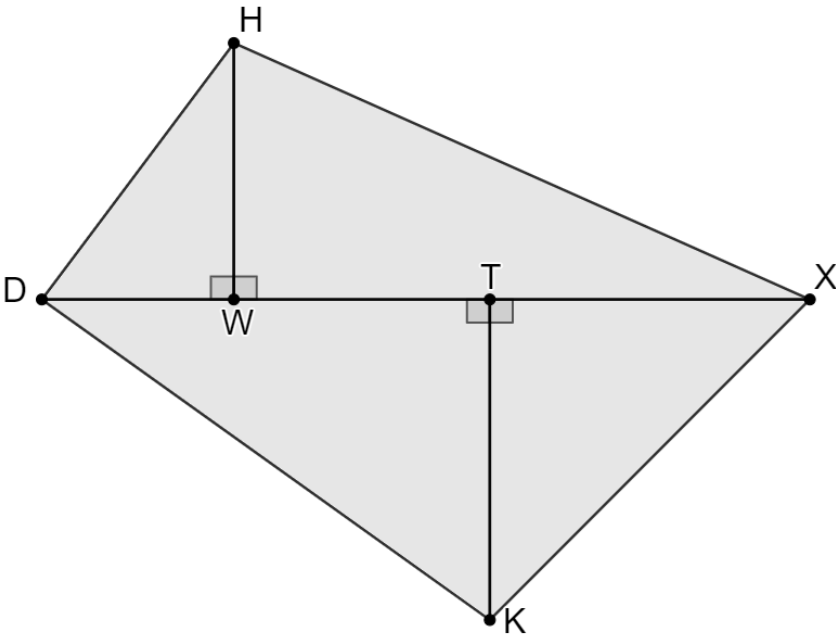
Right triangles YTN , RNY , and NRB are shown where $BR = 6$, $m\angle BNR = 48^\circ$ and $m\angle NRY = 29^\circ$.



Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure.

	Side	Work	Side Measure
1.	$\overline{NB}$	$\tan 48 = \frac{6}{NB}$	5.402
2.	$\overline{NR}$	$\sin 48 = \frac{6}{NR}$	8.074
3.	$\overline{YR}$	$\cos 29 = \frac{8.074}{YR}$	9.231
4.	$\overline{YN}$	$\tan 29 = \frac{YN}{8.074}$	4.475
5.	$\overline{YT}$	$\sin 42 = \frac{YT}{4.475}$	2.994

Quadrilateral $HXKD$ is composed of right triangles DHW , WHX , TXK , and DTK . $WH = 4$, $TK = 5$, $m\angle WDH = 53^\circ$, $m\angle WHX = 66^\circ$, $m\angle TKX = 45^\circ$, and $m\angle TDK = 35^\circ$.



Complete the table. In the work column, write the equation or trig ratio used to determine the side measure

	Side	Work	Side Measure
6.	$\overline{HX}$	$\cos 66 = \frac{4}{HX}$	9.834
7.	$\overline{XK}$	$\cos 45 = \frac{5}{XK}$	7.071
8.	$\overline{DK}$	$\sin 35 = \frac{5}{DK}$	8.717
9.	$\overline{HD}$	$\sin 53 = \frac{4}{HD}$	5.009

10. What is the perimeter of quadrilateral $HXKD$?

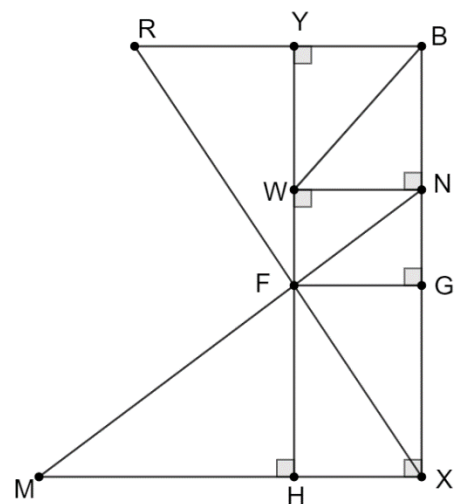
30.631

A pastry chef is creating a sculpture using chocolate. To ensure the sculpture has a strong base the chef uses a computer program to outline the structure he will use. The outline of the structure is shown.

The table shows known measurements of the structure.

$m\angle FMH = 37^\circ$	$FH = 4$ feet	$GN = 2$ feet
$m\angle XFH = 34^\circ$	$BN = 3$ feet	$m\angle YRF = 57^\circ$

Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure.



	Side	Work	Side Measure
1.	$\overline{HX}$	$\tan 34 = \frac{HX}{4}$	2.7
2.	$\overline{FM}$	$\sin 37 = \frac{4}{FX}$	6.6
3.	$\overline{RB}$	$\tan 57 = \frac{3 + 2 + 4}{RB}$	5.8

4. The polygon $FMXBR$ is the base of the chocolate structure. What is the area of the base?

$$MH: \tan 37 = \frac{4}{MH}$$

$$MH = 5.3$$

$$\triangle RBX: A = \frac{1}{2} \cdot 9 \cdot 5.8 = 26.1$$

$$\triangle MFH: A = \frac{1}{2} \cdot 5.3 \cdot 4 = 10.6$$

$$\triangle XFH: A = \frac{1}{2} \cdot 2.7 \cdot 4 = 5.4$$

$$26.1 + 10.6 + 5.4 = 42.1$$

Rounding of intermediate steps may result in slight answer variation.

The diagram shows a staircase that consists of 6 steps. The length of each step is 11 inches (in.) and the height each step is 5.25 in.

5. Determine the measure of $\angle CLR$. Round to the nearest hundredth.

$$\tan x = \frac{5.25}{11}; \tan^{-1} \frac{5.25}{11} = 25.51^\circ$$

6. Determine the length of $\overline{LW}$. Round to the nearest hundredth.

$$LW = 6 \cdot \sqrt{5.25^2 + 11^2} = 73.13$$

7. Determine the measure of $\angle HWT$. Round to the nearest hundredth.

$$\angle HWT \cong \angle CLR \text{ (alternate interior angles)}$$

$$m\angle HWT = 25.51^\circ$$

8. The homeowner plans to paint quadrilateral $HLNT$ using chalkboard paint. What is the area of quadrilateral $HLNT$? Round to the nearest hundredth.

$$A = \frac{1}{2} \cdot 22 \cdot (31.5 + 21) = 577.50$$

The crown of an oak tree is the area that consists of the branches and leaves of the tree. Leigh Ann was trying to determine the height of an oak tree's crown, TF . The height to Leigh Ann's eye, PW , is 5.5 feet (ft). Leigh Ann is standing 9 feet away from the tree. The angle of elevation from Leigh Ann's eye to the bottom of the oak tree's crown is 34° . The angle of elevation from Leigh Ann's eye to the top of the oak tree's crown is 45° .

9. Determine the height, in feet, of the tree. Round to the nearest tenth.

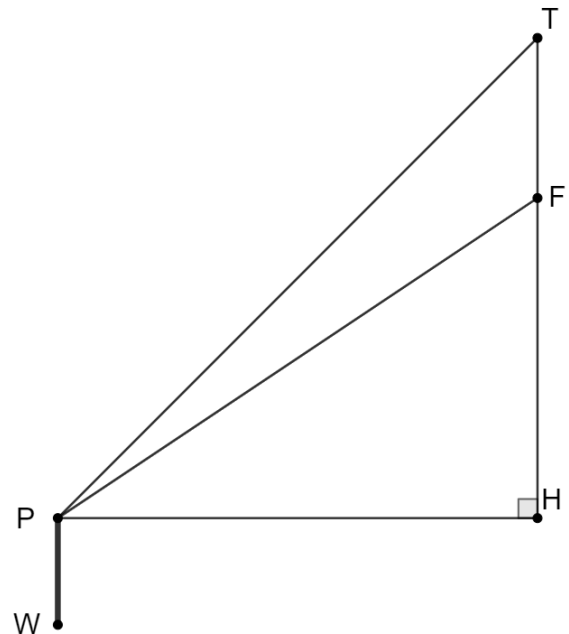
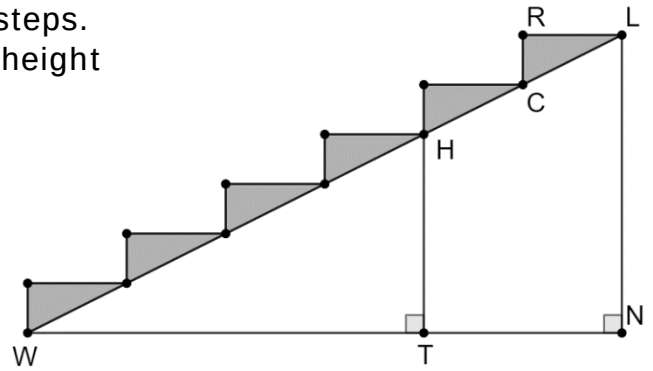
$$TH: \tan 45 = \frac{x}{9}; x = 9$$

$$9 + 5.5 = 14.5 \text{ ft}$$

10. Determine the height, in feet, of the oak tree's crown. Round to the nearest tenth.

$$FH: \tan 34 = \frac{x}{9}; x = 6.07$$

$$14.5 - 6.07 = 8.43 \text{ ft}$$



In $\triangle HRW$, $m\angle RHW = 32^\circ$, $m\angle HWR = 17^\circ$, and $WR = 25$. Determine the measurement of each part of $\triangle HRW$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
1.	$m\angle WRH$	131°
2.	$\overline{HR}$	13.8
3.	$\overline{HW}$	35.6

1) $180 - 32 - 17 = 131$

2) $\frac{\sin 32}{25} = \frac{\sin 17}{HR}$; $25 \sin 17 = HR(\sin 32)$; $\frac{25 \sin 17}{\sin 32} = HR$; $13.8 = HR$

3) $\frac{\sin 32}{25} = \frac{\sin 131}{HW}$; $25 \sin 131 = HW(\sin 32)$; $\frac{25 \sin 131}{\sin 32} = HW$; $HW = 35.6$

In $\triangle BTX$, $m\angle TXB = 26^\circ$, $TB = 8$ and $XT = 11.2$. Determine the measurement of each part of $\triangle BTX$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
4.	$m\angle XBT$	37.9°
5.	$m\angle XTB$	116.1°
6.	$\overline{XB}$	16.4

4) $\frac{\sin 26}{8} = \frac{\sin \theta}{11.2}$; $11.2 \sin 26 = 8 \sin \theta$; $\sin^{-1} 0.6137 = \theta$; $\theta = 37.9$

5) $180 - 26 - 37.9 = 116.1$

6) $\frac{\sin 26}{8} = \frac{\sin 116.1}{XB}$; $8 \sin 116.1 = XB \sin 26$; $XB = 16.4$

Shevrin and Tracey are both hiking in the Ocala National Forest. They are currently 12 miles apart. They plan to meet up at their campsite.

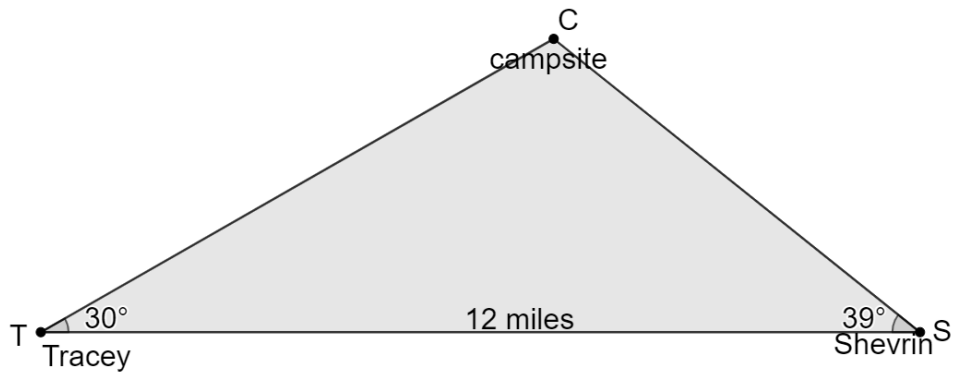
7. How far, to the nearest hundredth of a mile, does Tracey hike?

$$\frac{\sin 39}{x} = \frac{\sin 111}{12}$$

$$x \sin 111 = 12 \sin 39$$

$$x = \frac{12 \sin 39}{\sin 111}$$

$$x = 8.09 \text{ miles}$$



8. How far, to the nearest hundredth of a mile, does Shevrin hike?

$$\frac{\sin 30}{x} = \frac{\sin 111}{12}$$

$$x \sin 111 = 12 \sin 30$$

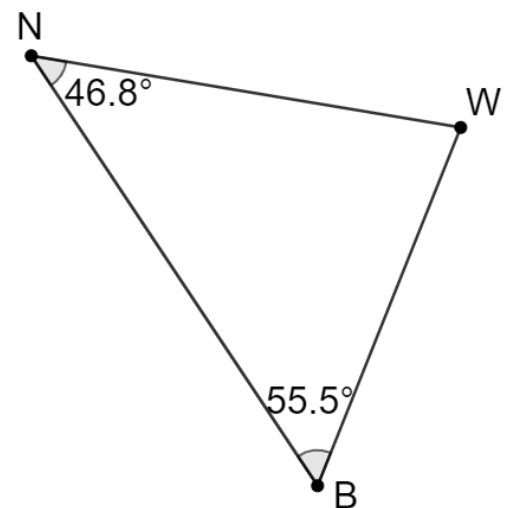
$$x = \frac{12 \sin 30}{\sin 111}$$

$$x = 6.43 \text{ miles}$$

Alejandro flies a drone in a competitive obstacle course. The obstacle course and some of its measurements are shown. The distance between point N and point B is 7.2 kilometers.

9. Write an equation that can be used to determine the distance, in kilometers, between point N and point W .

$$\frac{\sin 77.7}{7.2} = \frac{\sin 55.5}{x}$$



10. Determine the perimeter, in kilometers, of the obstacle course.

$$x \sin 77.7 = 7.2 \sin 55.5; x = \frac{7.2 \sin 55.5}{\sin 77.7}; x = 6.07 \text{ (NW)}$$

$$\frac{\sin 77.7}{7.2} = \frac{\sin 46.8}{x}; x \sin 77.7 = 7.2 \sin 46.8; x = \frac{7.2 \sin 46.8}{\sin 77.7}; x = 5.27 \text{ (WB)}$$

$$7.2 + 6.07 + 5.27 = 18.54 \text{ km}$$

Geometry

Unit 7

Lesson H2 Homework

Name Answer Key

Date _____

Period _____

In $\triangle TBN$, $m\angle TNB = 46^\circ$, $TN = 17.2$, and $BN = 19.4$. Determine the measurement of each part of $\triangle TBN$. Show your work below the table. If necessary, round to the nearest tenth.

Part	Measurement
1. $\angle NTB$	75.2°
2. $\angle TBN$	59°
3. $\overline{BT}$	14.4

$$\begin{aligned}
 1) \quad 19.4^2 &= 14.4^2 + 17.2^2 - 2(14.4)(17.2) \cos T & 2) \quad 17.2^2 &= 14.4^2 + 19.4^2 - 2(14.4)(19.4) \cos B \\
 376.36 &= 207.36 + 295.84 - 495.36 \cos T & 295.84 &= 207.36 + 376.36 - 558.72 \cos B \\
 -126.84 &= -495.36 \cos T & -287.88 &= -558.72 \cos B \\
 0.256 &= \cos T & 0.515 &= \cos B \\
 T &= 75.2^\circ & B &= 59^\circ
 \end{aligned}$$

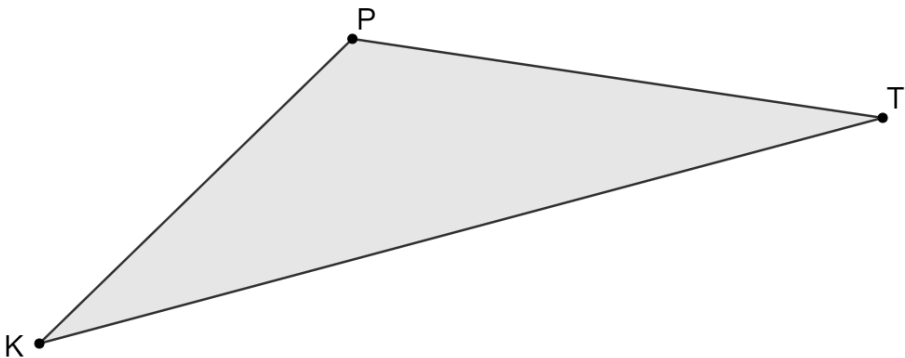
$$\begin{aligned}
 3) \quad BT^2 &= 19.4^2 + 17.2^2 - 2(19.4)(17.2) \cos 46 = 208.61 \\
 BT &= 14.4
 \end{aligned}$$

In $\triangle KRM$, $KR = 5$, $MK = 5$, and $MR = 7.826$. Determine the measurement of each part of $\triangle KRM$. Show your work below the table. If necessary, round to the nearest tenth.

Part	Measurement
4. $\angle RKM$	103°
5. $\angle MRK$	38.5°
6. $\angle KMR$	38.5°

$$\begin{aligned}
 5^2 &= 7.826^2 + 5^1 - 2(7.826)(5)(\cos R) \\
 25 &= 61.246 + 25 - 78.26 \cos R \\
 -61.246 &= -78.26 \cos R \\
 0.783 &= \cos R \\
 \angle R &= 38.5^\circ, \quad \angle M = 38.5^\circ \\
 \angle K &= 180 - 2(38.5) = 103^\circ
 \end{aligned}$$

When Camila visits the zoo she walks from the kangaroos, point K , to the Pandas, point P , then to the tigers, point T , before returning to the Kangaroos. The path from the kangaroos to the pandas is 398 yards long. The path from the pandas to the tigers is 488 yards long. The path from the tigers to the kangaroos is 795 yards long. Determine each angle measure.



	Angle	Measurement
7.	$\angle KPT$	127.3°
8.	$\angle PTK$	23.5°
9.	$\angle TKP$	29.2°

10. In $\triangle CYL$, $CY = 193$, $LY = 95$, and $CL = 168$. Complete the statement about $\triangle CYL$.

$\triangle CYL$ is a

☒ right
☐ acute
☐ obtuse
☐ right

triangle because

☒ one of the angles is
☐ all of the angles are

☒ equal to 90°.
☐ less than 90°.
☐ greater than 90°.

Determine the area of each triangle using the given information. Round your answer to the nearest hundredth.

	Triangle	Given Information			Area
1.	$\triangle HTR$	$HR = 15$	$TR = 19$	$m\angle HRT = 32^\circ$	$\frac{1}{2}(15)(19) \sin 32$ $= 75.51$
2.	$\triangle FNW$	$NW = 23$	$FN = 31$	$m\angle FNW = 107^\circ$	$\frac{1}{2}(23)(31) \sin 107$ $= 340.92$
3.	$\triangle BTM$	$BT = 3.9$	$MB = 2.7$	$m\angle MBT = 56^\circ$	$\frac{1}{2}(3.9)(2.7) \sin 56$ $= 4.36$
4.	$\triangle DLX$	$DX = \sqrt{27}$	$LX = \sqrt{20}$	$m\angle DXL = 63^\circ$	$\frac{1}{2}(\sqrt{27})(\sqrt{20}) \sin 63$ $= 10.35$
5.	$\triangle GZW$	$ZW = \frac{65}{8}$	$GW = \frac{93}{10}$	$m\angle ZWG = 77^\circ$	$\frac{1}{2}\left(\frac{65}{8}\right)\left(\frac{93}{10}\right) \sin 77$ $= 36.81$

6.	$\triangle TMR$	$TR = 16.2$	$MT = 12.1$	$m\angle MTR = 51^\circ$	$\frac{1}{2}(16.2)(12.1) \sin 51$ $= 76.17$
7.	$\triangle CNW$	$NW = \sqrt{7}$	$CN = \sqrt{7}$	$m\angle WNC = 40^\circ$	$\frac{1}{2}(\sqrt{7})(\sqrt{7}) \sin 40$ $= 2.25$
8.	$\triangle BHX$	$HX = 9.8$	$XB = 11.1$	$m\angle HXB = 84^\circ$	$\frac{1}{2}(9.8)(11.1) \sin 84$ $= 54.09$
9.	$\triangle RTW$	$TW = 15$	$WR = 19$	$m\angle TWR = 145^\circ$	$\frac{1}{2}(15)(19) \sin 145$ $= 81.73$
10.	$\triangle XYZ$	$XY = 4\sqrt{3}$	$YZ = 3\sqrt{2}$	$m\angle XYZ = 129^\circ$	$\frac{1}{2}(4\sqrt{3})(3\sqrt{2}) \sin 129$ $= 11.42$